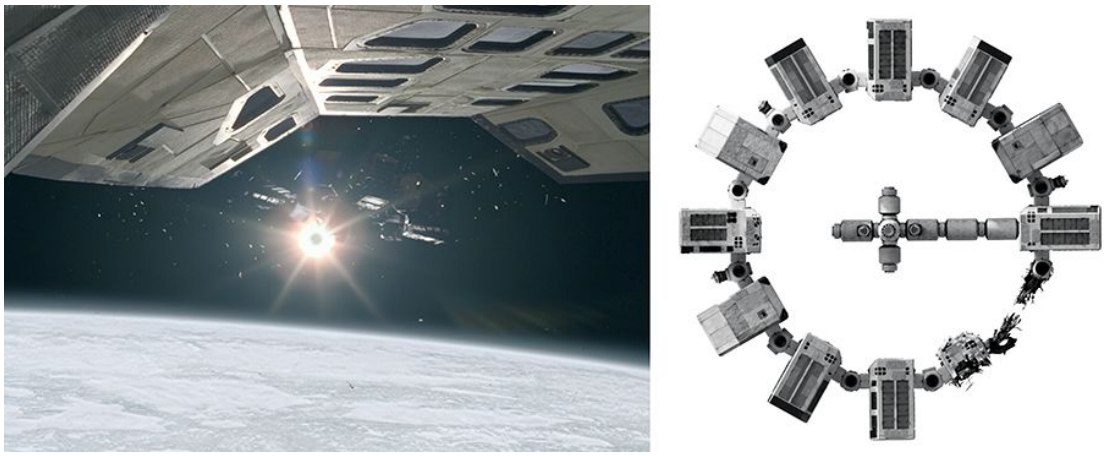


Fig. 20.2. *Left*: The explosion on the *Endurance*, with the lander above and Mann's planet below. (The ten radial light beams are lens flare due to scattering of light in the camera lens, not stuff from the explosion.) *Right*: The damaged *Endurance* after the explosion. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

The explosion sets the ring spinning so fast that its modules feel 70 gees (70 Earth gravities) of centrifugal force. Its broken ends swing apart from each other but don't break, and the ring's modules don't crash into each other. This, in my scientist's interpretation, is a great example of conservative design by clever engineers!

Incidentally, I'm impressed by the explosion in the movie. An explosion in space makes no sound, as there is no air to transmit the sound waves. The *Endurance* explosion has no sound. The flames in such an explosion must quench quickly, as the oxygen that feeds them is quickly disbursed into space. The flames in the movie quench quickly. Paul Franklin tells me that his team worked hard to achieve this, as the explosion was a real one, on a movie set, and not a computer-generated visual effect. Another example of Christopher Nolan's commitment to science accuracy.